PES UNIVERSITY, BANGALORE
Department of Computer Science and Engineering

SOFTWARE REQUIREMENTS SPECIFICATION

Dehallucination Add-On Module for Enhancing Reliability of AI Applications

Project ID: 39

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Version: 1.0 **Status:** Draft for Review **Date:** 06-09-2026

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Revision History

Version	Date	Author	Change Summary
1.0	06-09-2026	Team 39	Initial SRS covering requirements, interfaces, security, models and traceability.

1. Introduction

1.1 Purpose

This Software Requirements Specification defines the functional, non-functional, interface, security, data and verification requirements for a reusable dehallucination add-on module. The module receives AI-generated text, identifies factual claims, cross-references them with configured evidence sources or knowledge graphs, detects unsupported or contradictory claims, and produces a grounded response with confidence information.

1.2 Scope

Project ID 39 specifies a module that detects and reduces hallucinations in AI-generated text by cross-referencing facts with trusted sources or knowledge graphs and integrating with chatbots or content generators to improve accuracy.

• In scope: response ingestion, claim extraction, normalization, evidence retrieval, source assessment, verification, contradiction detection, confidence/risk scoring, grounded response generation, citations, reporting and audit logging.

• Out of scope: training a foundation model from scratch, unrestricted crawling, absolute guarantees of truth, and autonomous publication of corrections in high-stakes situations without configured review.

1.3 Intended Audience

Developers, QA engineers, application integrators, reviewers, administrators and project evaluators.

1.4 Definitions

Claim — factual proposition extracted from generated content. **Evidence** — information used to support, refute or qualify a claim. **Grounded response** — response whose factual statements are linked to evidence. **Hallucination** — unsupported, contradicted or unreliable AI-generated factual content relative to available evidence. **Trusted source** — source satisfying configured credibility rules.

2. Overall Description

2.1 Product Perspective

The product is middleware between a host AI application and its user or downstream content system. The host submits generated content; the module verifies claims and returns verdicts, evidence, confidence/risk information and an optional grounded response.

2.2 Major Product Functions

• Accept AI-generated text and optional query/domain/source-policy metadata.

• Extract and normalize factual claims.

• Retrieve supporting and contradicting evidence.

• Evaluate source relevance and configured credibility.

• Classify claims as Supported, Contradicted, Uncertain or Unverifiable.

• Calculate claim confidence and overall response risk.

• Generate grounded/corrected responses with evidence references.

• Persist verification events and produce reports.

2.3 User Classes

User Class	Characteristics / Needs
End User	Needs evidence indicators, citations and uncertainty warnings.
Application Developer	Integrates the add-on, configures policies and consumes the API.
Reviewer / Evaluator	Inspects claims, evidence, contradictions, scores and reports.
Administrator	Maintains trusted-source policy, access control and configuration.

2.4 Operating Environment

The prototype shall be deployable on a standard development workstation or containerized server with a browser/API client and network access when external evidence retrieval is enabled. Exact runtime and database technology are TBD at design stage.

2.5 Constraints and Assumptions

• Architecture shall be modular and replaceable at component boundaries.

• External sources may fail, rate-limit or provide incomplete evidence; the module must fail safely.

• No source access control or terms-of-service restriction may be bypassed.

• Each of the three team members owns a complete functionality.

• Verification quality depends on configured evidence sources and their freshness.

3. External Interface Requirements

3.1 User Interface

A web dashboard or developer interface shall display the submitted response, extracted claims, evidence references, verdicts, confidence/risk values, corrections, warnings and audit metadata. Error states shall be explicit and shall not be represented as successful verification.

3.2 Software Interfaces

The module interfaces with a host AI application API, evidence-provider interface, knowledge-graph interface and persistence layer. JSON is used for request and response payloads.

3.3 Communications

HTTPS/HTTP shall be used for service communication. Authentication tokens, when required, shall be transmitted only over encrypted transport. Upstream timeouts and failures shall produce machine-readable error states.

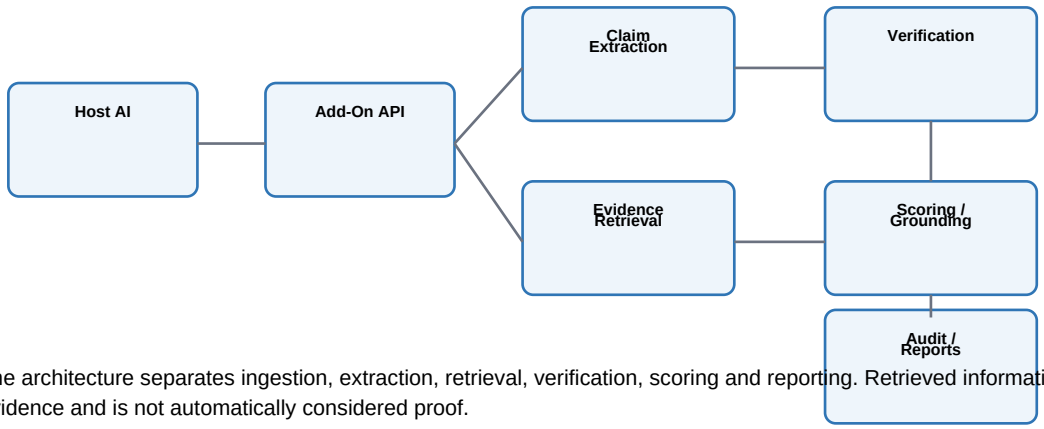
3.4 Proposed API Contract

Operation	Method / Path	Input	Output
Verify	POST /api/v1/verify	text, query?, domain?, policy?, mode?	claims, evidence, verdict, risk, grounded_response, warnings
Report	GET /api/v1/reports/{id}	report ID	verification report + audit metadata
Health	GET /api/v1/health	none	service/dependency status

4. Analysis Models

4.1 Proposed Architecture

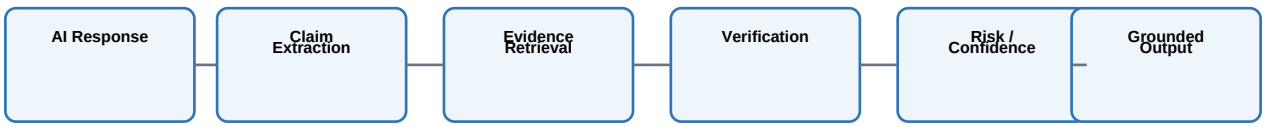
Proposed System Architecture



The architecture separates ingestion, extraction, retrieval, verification, scoring and reporting. Retrieved information is treated as evidence and is not automatically considered proof.

4.2 Main Verification Flow

Claim Verification Workflow



Missing or conflicting evidence produces an explicit uncertain/unverifiable outcome rather than an unsupported positive verdict.

5. System Features (Detailed)

Each functional requirement is uniquely identified and paired with a measurable acceptance criterion and test reference.

5.1 Request Ingestion & Validation

Priority: High

ID	Requirement	Stakeholder	Acceptance Criteria / Test
DH-F-001	Accept AI-generated response text.	Application Developer	Valid input creates request ID; invalid input is rejected. **TC-ING-01**
DH-F-002	Validate query, domain, source-policy and verification-mode parameters.	Application Developer	Invalid values are rejected before processing. **TC-ING-02**
DH-F-003	Assign a unique request ID and timestamp to each accepted request.	Reviewer	IDs are unique and timestamped. **TC-ING-03**

5.2 Claim Extraction & Normalization

Priority: High

ID	Requirement	Stakeholder	Acceptance Criteria / Test
DH-F-010	Identify factual claims and assign a unique claim ID to each.	Application Developer	Multiple statements produce separate claim records. **TC-CLM-01**
DH-F-011	Normalize extracted claims while retaining original claim text.	Application Developer	Original and normalized forms are retained. **TC-CLM-02**
DH-F-012	Classify claims as factual, numerical, temporal or entity-relation where applicable.	Reviewer	Representative claims receive expected classes. **TC-CLM-03**

5.3 Evidence Retrieval & Source Assessment

Priority: High

ID	Requirement	Stakeholder	Acceptance Criteria / Test
DH-F-020	Retrieve evidence from configured trusted sources and/or knowledge graphs.	Application Developer	Known test claims return relevant evidence. **TC-EVD-01**
DH-F-021	Store source identifier, label and retrieval timestamp for evidence.	Reviewer	Every evidence item contains required metadata. **TC-EVD-02**
DH-F-022	Rank evidence using configured relevance and source-credibility signals.	Administrator	Controlled policy changes alter ranking as expected. **TC-EVD-03**
DH-F-023	Report evidence-unavailable when acceptable evidence is not found.	End User	No-evidence case is explicitly Unverifiable/Uncertain. **TC-EVD-04**

5.4 Verification & Contradiction Detection

Priority: High

ID	Requirement	Stakeholder	Acceptance Criteria / Test
DH-F-030	Assign Supported, Contradicted, Uncertain or Unverifiable verdicts.	Reviewer	Controlled cases receive expected verdicts. **TC-VER-01**
DH-F-031	Detect material contradictions between claims and evidence.	Reviewer	Contradictory pair is flagged with evidence references. **TC-VER-02**
DH-F-032	Preserve conflicting evidence instead of silently discarding disagreement.	Reviewer	Conflicting evidence is visible with a conflict indicator. **TC-VER-03**
DH-F-033	Prevent Supported verdict when minimum evidence conditions are not met.	Application Developer	Insufficient-evidence cases remain uncertain/unverifiable. **TC-VER-04**

5.5 Confidence & Risk Scoring

Priority: High

ID	Requirement	Stakeholder	Acceptance Criteria / Test
DH-F-040	Calculate a confidence score for each claim using evidence and consistency signals.	Reviewer	Score is within configured range and returned with verdict. **TC-SCR-01**

ID	Requirement	Stakeholder	Acceptance Criteria / Test
DH-F-041	Calculate an overall response risk indicator from claim-level results.	End User	Contradicted high-impact content produces higher risk. **TC-SCR-02**
DH-F-042	Expose reasons contributing to low confidence or high risk.	Reviewer	Report identifies evidence gaps, contradictions or source issues. **TC-SCR-03**

5.6 Grounded Response Generation

Priority: High

ID	Requirement	Stakeholder	Acceptance Criteria / Test
DH-F-050	Generate a grounded response that revises or qualifies uncertain/contradicted claims when enabled.	End User	Controlled hallucination cases produce corrected/qualified output. **TC-GRD-01**
DH-F-051	Attach evidence references to factual statements where evidence exists.	End User	Supported statements contain evidence references. **TC-GRD-02**
DH-F-052	Clearly indicate when a claim could not be verified.	End User	Unverifiable claims receive an uncertainty warning. **TC-GRD-03**

5.7 Reporting & Audit

Priority: Medium

ID	Requirement	Stakeholder	Acceptance Criteria / Test
DH-F-060	Generate a verification report containing claims, verdicts, scores, evidence and warnings.	Reviewer	Completed report contains mandatory sections. **TC-REP-01**
DH-F-061	Persist an audit event for each request and final result.	Administrator	Each accepted request has a matching audit record. **TC-REP-02**
DH-F-062	Allow authorized users to retrieve a report by request ID.	Reviewer	Valid ID returns correct report; unknown ID returns controlled not-found. **TC-REP-03**

5.8 Configuration & Administration

Priority: Medium

ID	Requirement	Stakeholder	Acceptance Criteria / Test
DH-F-070	Allow authorized administrators to configure source classes, priorities and thresholds.	Administrator	Validated changes persist and affect subsequent requests. **TC-CFG-01**
DH-F-071	Provide health/status information for core services and evidence dependencies.	Administrator	Unavailable dependencies are accurately reported. **TC-CFG-02**

6. Non-Functional Requirements

ID	Requirement	Category	Priority	Acceptance
DH-NF-001	90% of normal requests containing up to 20 claims shall complete within 10 seconds, excluding upstream timeout delays.	Performance	High	90th percentile ≤10s; TC-PERF-01
DH-NF-002	Evidence dependencies shall time out after a configured limit and return a controlled error state.	Reliability	High	No indefinite blocking; TC-PERF-02
DH-NF-003	Independent claims shall continue processing when one retrieval operation fails.	Robustness	High	Partial result + warning; TC-ROB-01
DH-NF-004	Verification workers shall be deployable as independent/stateless instances.	Scalability	Medium	Two workers process independent requests; design review
DH-NF-005	Ingestion, extraction, retrieval, verification, scoring and reporting shall be separated into documented modules.	Maintainability	Medium	Module review; TC-MNT-01
DH-NF-006	A new integrator shall submit a sample request using API documentation within 15 minutes after setup.	Usability	Medium	3-user usability test; TC-UX-01
DH-NF-007	All payloads shall use UTF-8 and preserve Unicode text.	Interoperability	Medium	Multilingual round-trip; TC-INT-01
DH-NF-008	Repeated identical input/configuration shall preserve deterministic rule-based verdict components.	Correctness	Medium	Five-run consistency; TC-COR-01

6.1 Safety Requirements

• Retrieval success alone shall never be treated as proof.

• Missing, conflicting or stale evidence shall produce explicit uncertainty.

• High-impact configured domains shall support review-required mode before automatic correction/publication.

6.2 Security Requirements

ID	Requirement	Priority	Acceptance
DH-SR-001	Authenticate protected API/admin operations.	High	Unauthorized request is rejected; TC-SEC-01
DH-SR-002	Enforce role-based authorization for configuration and protected reports.	High	Out-of-role operation denied and logged; TC-SEC-02
DH-SR-003	Do not store credentials/secrets in plaintext in source code or repository files.	High	Repository/config review; TC-SEC-03
DH-SR-004	Log security-relevant administrative changes with actor, time, action and outcome.	High	Audit test passes; TC-SEC-04
DH-SR-005	Use HTTPS/TLS for deployed network communication.	High	Transport inspection; TC-SEC-05
DH-SR-006	Provide configurable retention/deletion for stored verification requests.	Medium	Retention test passes; TC-SEC-06

7. Quality Attributes & Acceptance Tests

• Correctness: verdicts must be evidence-dependent and traceable.

• Reliability: dependency failures degrade to explicit uncertainty.

• Maintainability: retrieval and verification strategies are replaceable through interfaces.

• Testability: claim processing can be tested independently of the UI.

• Interoperability: JSON/HTTPS and UTF-8 support host applications.

• Usability: users can understand why a claim was flagged and what evidence was used.

7.1 Acceptance Exit Criteria

All High-priority functional requirements must be implemented and verified; no critical security failure may remain; NFR measurements must be recorded; and every listed requirement must have a mapped test case in the RTM.

8. System Models and Diagrams

The architecture and verification workflow in Section 4 are the principal proposed models. The separation between evidence retrieval and verification allows evidence to be inspected before a verdict is assigned.

9. Other Requirements

9.1 Database Requirements

Logical entities shall include VerificationRequest, Claim, Evidence, Source, VerificationResult, Configuration and AuditEvent. Unique identifiers and queryable claim-to-evidence relationships are required for report generation.

9.2 Internationalization

All interfaces shall use UTF-8. Unicode text must survive ingestion, storage and response processing without corruption.

9.3 Legal, Ethical and Privacy

• The module is not an absolute truth oracle; results depend on configured evidence.

• User content shall be processed for the stated verification purpose and retained according to policy.

• Source access must respect applicable licenses, access controls and terms.

• High-stakes use should support human review and explicit uncertainty.

9.4 Business / Domain Rules

• Retrieved content is evidence, not automatically proof.

• Claims without acceptable evidence cannot be labelled Supported.

• Contradictory evidence must lower confidence and be explained.

• Corrections must not introduce unsupported factual assertions.

• Configuration changes apply to subsequent requests unless a report is explicitly re-run.

9.5 Team Functionality Ownership

Member	Owned Functionality	Requirements
Abhinav Chugh	Claim extraction, normalization and verification	DH-F-010–033
Aarush Nalwaya	Evidence retrieval and trusted-source integration	DH-F-020–023, DH-F-070
Adarsh Teli	Confidence, grounded response and reporting	DH-F-040–062

Appendix A: Glossary

Term	Definition
AI	Artificial Intelligence
API	Application Programming Interface
Claim	Factual proposition extracted from generated content
Confidence Score	Numeric indication of evidential support
Dehallucination	Detection, reduction or correction of unsupported AI claims
Evidence	Information used to support, refute or qualify a claim
Grounded Response	Response whose factual statements are linked to evidence
Knowledge Graph	Structured entities, relations and attributes used as an evidence source
RBAC	Role-Based Access Control
Trusted Source	Configured source satisfying credibility policy
Unverifiable	Claim for which acceptable evidence was not obtained

Appendix B: Field Layouts & Report Requirements

Field	Length	Type	Description	Mandatory
request_id	36	UUID/String	Unique verification request identifier	Y
input_text	TBD	Text	AI-generated response	Y
query_context	TBD	Text	Optional query/context	N
domain	64	String	Optional domain	N
verification_mode	20	Enum	verify-only / grounded / review-required	Y
source_policy	64/TBD	String/JSON	Trusted-source policy	N
created_at	Timestamp	DateTime	Creation time	Y

Report Requirements

• Verification Summary: request ID, processing time, overall verdict/risk and warning count.

• Claim Verification: claim ID, original/normalized claim, verdict, confidence, evidence and contradiction references.

• Evidence Detail: source ID, label, source type, retrieval time and quality indicators.

• Grounded Response: grounded text, correction notes, citations and unresolved claims.

• Audit Record: actor/request ID, timestamp, action, outcome and configuration version.

Appendix C: Requirements Traceability Matrix

N = Not implemented at SRS stage. Architecture, design and code references are intentionally TBD until later phases.

ID	Brief Description	Architecture	Design	Code	Test	System Test	Status
DH-F-001	Accept AI-generated response text.	ARCH-TBD	DES-TBD	CODE-TBD	TC-ING-01	SYS-TBD	N
DH-F-002	Validate query, domain, source-policy and ver...	ARCH-TBD	DES-TBD	CODE-TBD	TC-ING-02	SYS-TBD	N
DH-F-003	Assign a unique request ID and timestamp to e...	ARCH-TBD	DES-TBD	CODE-TBD	TC-ING-03	SYS-TBD	N
DH-F-010	Identify factual claims and assign a unique c...	ARCH-TBD	DES-TBD	CODE-TBD	TC-CLM-01	SYS-TBD	N
DH-F-011	Normalize extracted claims while retaining or...	ARCH-TBD	DES-TBD	CODE-TBD	TC-CLM-02	SYS-TBD	N
DH-F-012	Classify claims as factual, numerical, tempor...	ARCH-TBD	DES-TBD	CODE-TBD	TC-CLM-03	SYS-TBD	N
DH-F-020	Retrieve evidence from configured trusted sou...	ARCH-TBD	DES-TBD	CODE-TBD	TC-EVD-01	SYS-TBD	N
DH-F-021	Store source identifier, label and retrieval ...	ARCH-TBD	DES-TBD	CODE-TBD	TC-EVD-02	SYS-TBD	N
DH-F-022	Rank evidence using configured relevance and ...	ARCH-TBD	DES-TBD	CODE-TBD	TC-EVD-03	SYS-TBD	N
DH-F-023	Report evidence-unavailable when acceptable e...	ARCH-TBD	DES-TBD	CODE-TBD	TC-EVD-04	SYS-TBD	N
DH-F-030	Assign Supported, Contradicted, Uncertain or ...	ARCH-TBD	DES-TBD	CODE-TBD	TC-VER-01	SYS-TBD	N
DH-F-031	Detect material contradictions between claims...	ARCH-TBD	DES-TBD	CODE-TBD	TC-VER-02	SYS-TBD	N
DH-F-032	Preserve conflicting evidence instead of sile...	ARCH-TBD	DES-TBD	CODE-TBD	TC-VER-03	SYS-TBD	N
DH-F-033	Prevent Supported verdict when minimum eviden...	ARCH-TBD	DES-TBD	CODE-TBD	TC-VER-04	SYS-TBD	N
DH-F-040	Calculate a confidence score for each claim u...	ARCH-TBD	DES-TBD	CODE-TBD	TC-SCR-01	SYS-TBD	N
DH-F-041	Calculate an overall response risk indicator ...	ARCH-TBD	DES-TBD	CODE-TBD	TC-SCR-02	SYS-TBD	N
DH-F-042	Expose reasons contributing to low confidence...	ARCH-TBD	DES-TBD	CODE-TBD	TC-SCR-03	SYS-TBD	N
DH-F-050	Generate a grounded response that revises or ...	ARCH-TBD	DES-TBD	CODE-TBD	TC-GRD-01	SYS-TBD	N
DH-F-051	Attach evidence references to factual stateme...	ARCH-TBD	DES-TBD	CODE-TBD	TC-GRD-02	SYS-TBD	N
DH-F-052	Clearly indicate when a claim could not be ve...	ARCH-TBD	DES-TBD	CODE-TBD	TC-GRD-03	SYS-TBD	N
DH-F-060	Generate a verification report containing cla...	ARCH-TBD	DES-TBD	CODE-TBD	TC-REP-01	SYS-TBD	N
DH-F-061	Persist an audit event for each request and f...	ARCH-TBD	DES-TBD	CODE-TBD	TC-REP-02	SYS-TBD	N
DH-F-062	Allow authorized users to retrieve a report b...	ARCH-TBD	DES-TBD	CODE-TBD	TC-REP-03	SYS-TBD	N
DH-F-070	Allow authorized administrators to configure ...	ARCH-TBD	DES-TBD	CODE-TBD	TC-CFG-01	SYS-TBD	N
DH-F-071	Provide health/status information for core se...	ARCH-TBD	DES-TBD	CODE-TBD	TC-CFG-02	SYS-TBD	N
DH-NF-001	90% of normal requests containing up to 20 cl...	ARCH-TBD	DES-TBD	CODE-TBD	90th percentile ≤10s; TC-PERF-01	SYS-TBD	N
DH-NF-002	Evidence dependencies shall time out after a ...	ARCH-TBD	DES-TBD	CODE-TBD	No indefinite blocking; TC-PERF-02	SYS-TBD	N
DH-NF-003	Independent claims shall continue processing ...	ARCH-TBD	DES-TBD	CODE-TBD	Partial result + warning; TC-ROB-01	SYS-TBD	N
DH-NF-004	Verification workers shall be deployable as i...	ARCH-TBD	DES-TBD	CODE-TBD	Two workers process independent requests; design review	SYS-TBD	N

ID	Brief Description	Architecture	Design	Code	Test	System Test	Status
DH-NF-005	Ingestion, extraction, retrieval, verificatio...	ARCH-TBD	DES-TBD	CODE-TBD	Module review; TC-MNT-01	SYS-TBD	N
DH-NF-006	A new integrator shall submit a sample reques...	ARCH-TBD	DES-TBD	CODE-TBD	3-user usability test; TC-UX-01	SYS-TBD	N
DH-NF-007	All payloads shall use UTF-8 and preserve Uni...	ARCH-TBD	DES-TBD	CODE-TBD	Multilingual round-trip; TC-INT-01	SYS-TBD	N
DH-NF-008	Repeated identical input/configuration shall ...	ARCH-TBD	DES-TBD	CODE-TBD	Five-run consistency; TC-COR-01	SYS-TBD	N
DH-SR-001	Authenticate protected API/admin operations.	ARCH-TBD	DES-TBD	CODE-TBD	Unauthorized request is rejected; TC-SEC-01	SYS-TBD	N
DH-SR-002	Enforce role-based authorization for configur...	ARCH-TBD	DES-TBD	CODE-TBD	Out-of-role operation denied and logged; TC-SEC-02	SYS-TBD	N
DH-SR-003	Do not store credentials/secrets in plaintext...	ARCH-TBD	DES-TBD	CODE-TBD	Repository/config review; TC-SEC-03	SYS-TBD	N
DH-SR-004	Log security-relevant administrative changes ...	ARCH-TBD	DES-TBD	CODE-TBD	Audit test passes; TC-SEC-04	SYS-TBD	N
DH-SR-005	Use HTTPS/TLS for deployed network communicat...	ARCH-TBD	DES-TBD	CODE-TBD	Transport inspection; TC-SEC-05	SYS-TBD	N
DH-SR-006	Provide configurable retention/deletion for s...	ARCH-TBD	DES-TBD	CODE-TBD	Retention test passes; TC-SEC-06	SYS-TBD	N

The RTM is intentionally marked N/TBD at requirements stage. It must be updated during architecture, implementation and validation; no future code or test evidence is fabricated in this document.